

表 9-5. BQ769x0 Design Requirements (continued)

DESIGN PARAMETER	EXAMPLE VALUE at T _A = 25°C
SCD Protection Delay Time	100 μs

9.2.2 Detailed Design Procedure

To begin the design process, there are some key steps required for component selection and protection configuration.

9.2.2.1 Step-by-Step Design Procedure

- Determine the number of series cells.
 - This value depends on the cell chemistry and the load requirements of the system. For example, to support a minimum battery voltage of 24 V using Li-CO₂ type cells with a cell minimum voltage of 3.0 V, there needs to be at least 8-series cells.
- Select the correct BQ769x0 device.
 - For 8-series cells, the BQ76930 is needed.
 - For the correct cell connections, see 表 9-3.
- Select the correct protection FETs.
 - The BQ76930 uses a low-side drive suitable for N-CH FETs.
 - These FETs should be rated for the maximum:
 - Voltage, which should be approximately 5 V (DC) 10 V (peak) per series cell: for example, 40 V.
 - Current, which should be calculated based on both the maximum DC current and the maximum transient current with some margin: for example, 30 A.
 - Power Dissipation, which can be a factor of the RDS(ON) rating of the FET, the FET package, and the PCB design: for example, 5 W, assuming 5 mΩ RDS(ON).
- Select the correct sense resistor.
 - The resistance value should be selected to maximize the input bandwidth use of the coulomb counter range, CC_{RANGE}, as well as keeping the SCD and OCD thresholds in the available selections, and not exceed the absolute maximum ratings. The sense resistance RSNS is the threshold or input voltage divided by the current.
 - Using the normal max discharge current, RSNS = 200 mV / 10.0 A = 20 mΩ maximum.
 - However, considering the maximum SCD threshold of 200 mV and I_{SCD} of 25 A, RSNS = 200 mV / 25 A = 8 mΩ maximum.
 - The maximum OCD threshold available is 100 mV, with the maximum current of 15 A, RSNS = 100 mV / 15.0 A = 6.7 mΩ maximum.
 - Further tolerance analysis (value tolerance, temperature variation, and so on) and PCB design margin should also be considered, so RSNS of 5 mΩ would be suitable with a 75-ppm temperature coefficient and power rating of 5 W.
 - With VSS referenced at the SRP terminal charge current creates a negative voltage on SRN. The 5 mΩ with 3 A charge current is within the absolute maximum range.
- The BQ76930 is chosen, and so the REGSRC pin needs to be powered through a source follower circuit where the FET is used to provide current for REGSRC from the battery positive terminal while reducing the voltage to a suitable value for the IC.
 - The FET also dissipates the power resulting from the load current and dropped voltage external to the IC and care should be taken to ensure the correct dissipation ratings are specified by the chosen FET.
- Configure the Current-based protection settings through PROTECT1 and PROTECT2:
 - Ideal SCD Threshold = 25 A × 5 mΩ = 125 mV.
 - However, the closest options are 111 mV (0x03) and 133 mV (0x04) providing 22.2 A and 26.6 A, respectively. Both options have the RSNS bit = 1.
 - 0x03 (22.2 A) will be used in this example.
 - The SCD delay threshold setting for a 100 μs delay is 0x01.
 - Therefore, PROTECT1 should be programmed with 0x8B.